

C-IMMSIM simulation results

January 5, 2026

Abstract

This document includes the plots relative to the simulation and the outcome of the epitope/peptide prediction used.

Produced by the C-IMMSIM Online server available at
<http://c-immsim.iac.rm.cnr.it> (alias to <http://kraken.iac.rm.cnr.it/C-IMMSIM>)

CITATIONS: For publication of results, please cite the following:

Nicolas Rapin, Ole Lund, Massimo Bernaschi, Filippo Castiglione. Computational Immunology Meets Bioinformatics: The Use of Prediction Tools for Molecular Binding in the Simulation of the Immune System. PLoS ONE 5(4): e9862
doi:10.1371/journal.pone.0009862, 2010

A retrospective validation

In-silico evaluation of adenoviral COVID-19 vaccination protocols: Assessment of immunological memory up to 6 months after the third dose. P. Stolfi, F. Castiglione, E. Mastrostefano, I. Di Biase, S. Di Biase, G. Palmieri, A. Prisco. Frontiers in Immunology, 13 (2022) doi: 10.3389/fimmu.2022.998262
<https://www.frontiersin.org/articles/10.3389/fimmu.2022.998262>

An in vivo validation

Identification and validation of viral antigens sharing sequence and structural homology with tumor associated antigens (TAAs). C. Ragone, C. Manolio, B. Cavalluzzo, A. Petrizzo, A. Mauriello, M-L. Tornesello, F. M. Buonaguro, F. Castiglione, L. Vitagliano, M. Ruvo, M. Tagliamonte, L. Buonaguro. Journal for ImmunoTherapy of Cancer. 9:e002694 (2021)
<https://jitc.bmj.com/content/9/5/e002694>

Original C-IMMSIM model: www.iac.cnr.it/~filippo/c-immsim

GETTING HELP:

Scientific problems: Filippo Castiglione (filippo dot castiglione at cnr dot it)

Technical problems: Ilaria Gonnella (ilaria dot gonnella at cnr dot it)

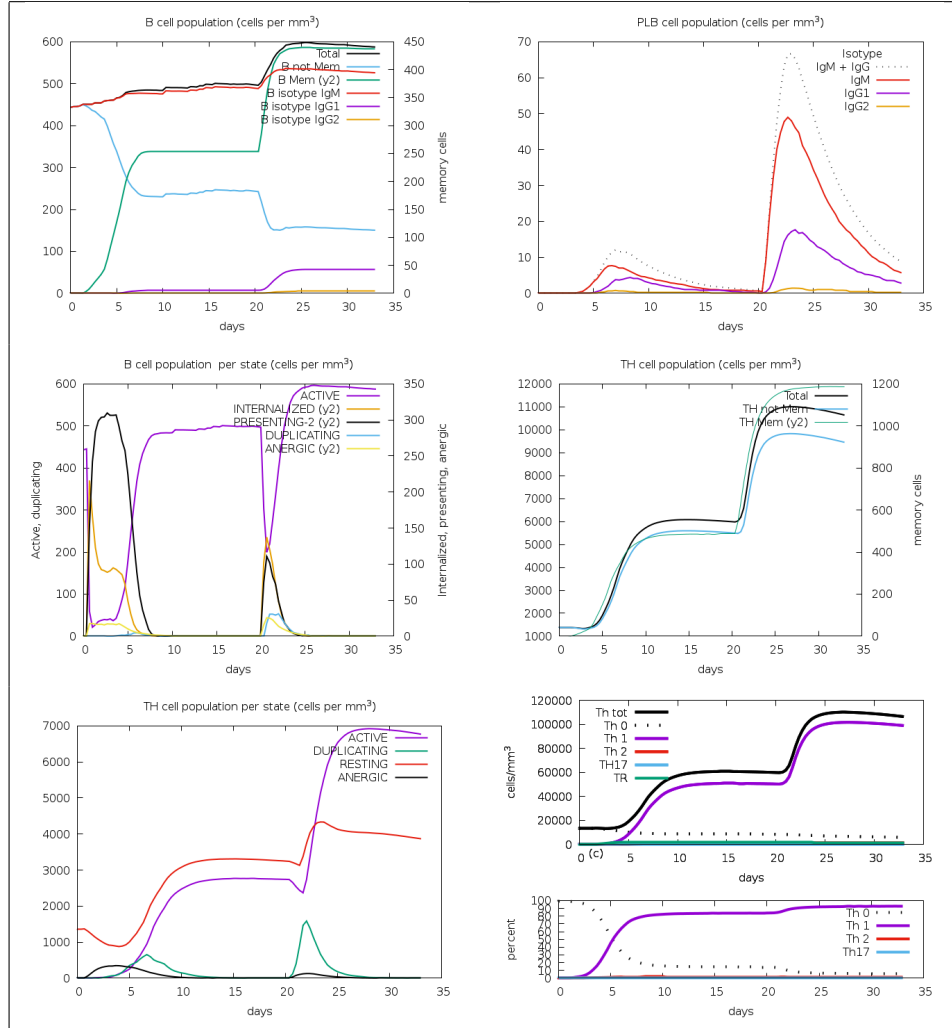


Figure 1: Cell counts shown. Legend: Act=active, Intern=internalized the Ag, Pres II = presenting on MHC II, Dup = in the mitotic cycle, Anergic = anergic, Resting = not active.

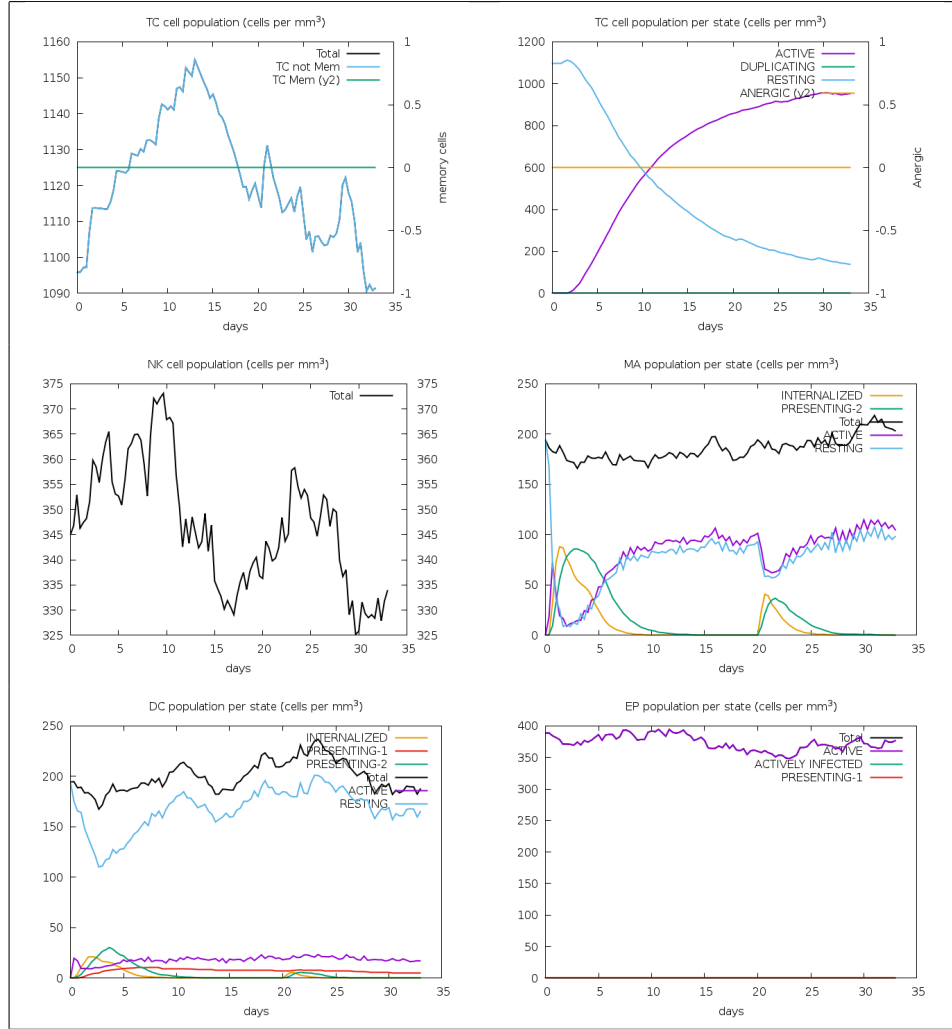


Figure 2: Legend: symbols as figure above.

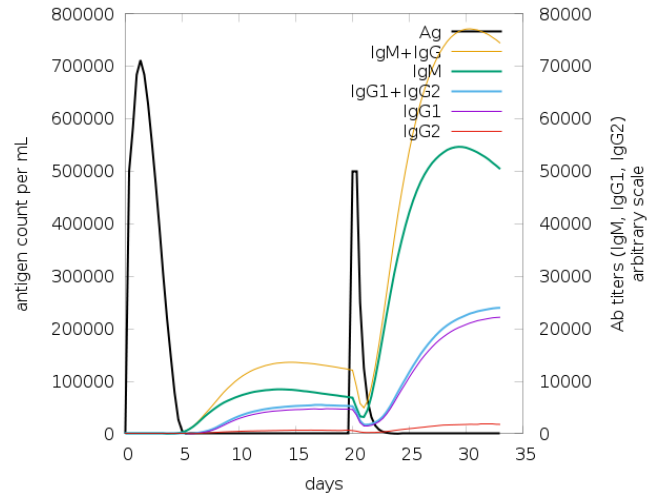


Figure 3: The virus, the immunoglobulins and the immunocomplexes.

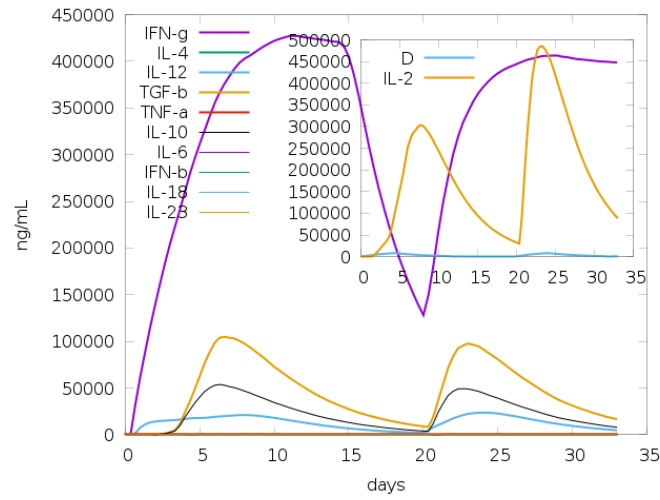


Figure 4: Concentration of cytokines and interleukins. Inset plot shows danger signal together with leukocyte growth factor IL-2.

Use Parker's propensity scale, takes an antigen block as input, and creates a list of residues that are possible epitopes.

APHALSGGSAPTGSWFSAPQAPENAYQAYAAPQLFPVSDITQNNQTNQAGGEAPQPGDNSTVQEAAAKGPGTGPNGNL
GEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGRRPGAPGSGSGSPRHRDGVRRPQKRPSICGCKGTHGGTGAGAG
AGGAGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAAAKRVLSVVLLGPGPGMTMFEV
SVAFGPGPGACMLSAPLEKGPGLPQGGQLTAYGPGPGFLDKGTYTLGPGPSLREDRDAWHLPAEAAA

```
00000000011100000000011000000000000000001111111111111111110001001111110001  
111111111111111111111111111111111111111111111111101110011100000001111111111  
11111111111111111111111111111111111111111111111111100000000000000000000000  
0000000000000000000010000000000000000000000000000000000011111000000000
```

```

1] pos=42 len=22      TQNQQTNQAGGEAPQPGDNSTV
2] pos=70 len=6       GPGTGP
3] pos=79 len=53      LGEKGDTSGPEGSGSGSPQRRGGDNHGRGRGRGRGRGGGRPGAPGSGSGSPRH
4] pos=133 len=4      DGVR
5] pos=149 len=66     GTHGGTGAGAGAGGAGAGGAGAGGGAGGAGGAGGAGAGGGAGGAGGAGAGGGAG
6] pos=294 len=6      EREDRD

```

Given the antigen injected creates the list of peptides for all the NumAgProts proteins and for all i.e., 4 MHCI molecules

Read class I peptide list from file? NO

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000

APPHALSGGSAPTGSWFSAPQAPENAYQAYAAAPQLFPVSDITQNNQTNQAGGEAPQPGDNSTVQEAALKPGTGPNGNL
GEKGDTSCPEGSGGSGPQRRGGDNHGRGRGRGRGRGGRRPGAPGSGSGSPRHRDGVRRPQKRPSICGCKGTHGGTGAGAG
AGGAGAGGAGAGGAGAGGAGGAGGAGGAGAGGAGAGGAGGAGGAGGAGGAGGAGGAGAAAKRVLSVVLLGPGPGMTMFEV
SVAFGPGPGACMLSAPEKGPGPGLPQGQLTAYGPGPGFLDKGTYTLGPGPGSLEREDRAWHLPAAEAAA

```
0] pos= -1 score=1.000000 unnormalised=104.8410000000 non-binding event
```

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-014700_5_105BEA2N1Swg.FSA_1_001

```
0] pos= -1 score=1.000000 unnormalised=104.8410000000 non-binding event
```

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-014700_5_105BEA2N1Swg.FSA_1_001

0]	pos= 10	score=0.021143	unnormalised=2.3742000000	APTSWFSFA
1]	pos= 18	score=0.002664	unnormalised=0.2992000000	APQPAPENA
2]	pos= 54	score=0.008711	unnormalised=0.9782000000	APQPGDNST
3]	pos= 118	score=0.002682	unnormalised=0.3012000000	RPQAPGGSG
4]	pos= 137	score=0.017207	unnormalised=1.9322000000	RPQKRPSCI
5]	pos= 261	score=0.013956	unnormalised=1.5672000000	GPGLPQGQL
6]	pos= -1	score=0.933636	unnormalised=104.8410000000	non-binding event

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-014700_5_105BEA2N1Swg.FSA_1_001

0]	pos= 10	score=0.021143	unnormalised=2.3742000000	APTGSWFSA
1]	pos= 18	score=0.002664	unnormalised=0.2992000000	APQPAPENA
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3]	pos= 118	score=0.002682	unnormalised=0.3012000000	RPGAPGGSG
4]	pos= 137	score=0.017207	unnormalised=1.9322000000	RPQKRPSCI
5]	pos= 261	score=0.013956	unnormalised=1.5672000000	GPGLPQQGL

7] pos= -1 score=0.757765 unnormalised=66.1680000000 non-binding event
